

Craftax runs compared: achievements and what unlocks them

12 finished runs, 1.1M env steps each, size50m, one training seed per run. Scored with tools/death_eval.py on the same fixed evaluation worlds. Updated 2026-10-05.

Every number is the mean episode-final achievement count over the evaluation episodes, and each unlock rate is the share of episodes in which that achievement fired. Every run was scored on the same fixed evaluation worlds; re-scoring two old checkpoints with today's code reproduced them exactly.

The two-level agent sets the record. Manager v1.1 -- B2 plus a manager that picks a tech-tree goal every 8 steps, with both levels paid when the goal is reached -- reaches 8.53 achievements, against 7.60 for B2 and 7.13 for manager v1, whose manager never learned to choose (it picked its 13 goals uniformly to the end). v1.1 places the furnace (67%) and stone (67%) far more often than B2 (40%, 40%), mines stone in 73% of episodes (B2 43%), and makes a stone sword (23%) and arrows (10%) for the first time. Its manager settled on the start of the tech tree in order: wood pickaxe, table, stone, wood sword. It still never makes the stone pickaxe -- the sword takes the ingredients -- and B2's coal (20%) and torches (3%) do not appear.

Experiment B before it: masking impossible actions (B2) 7.60 against 5.87 for the same model without it; validity flags as an input (B1) 6.90. The non-cheating model (Honest map) matches the privileged control (5.87 vs 5.93).

Every map run except the hand-written-rewards one unlocks WAKE_UP (sleep, then wake) in 80-98% of episodes; vanilla never does.

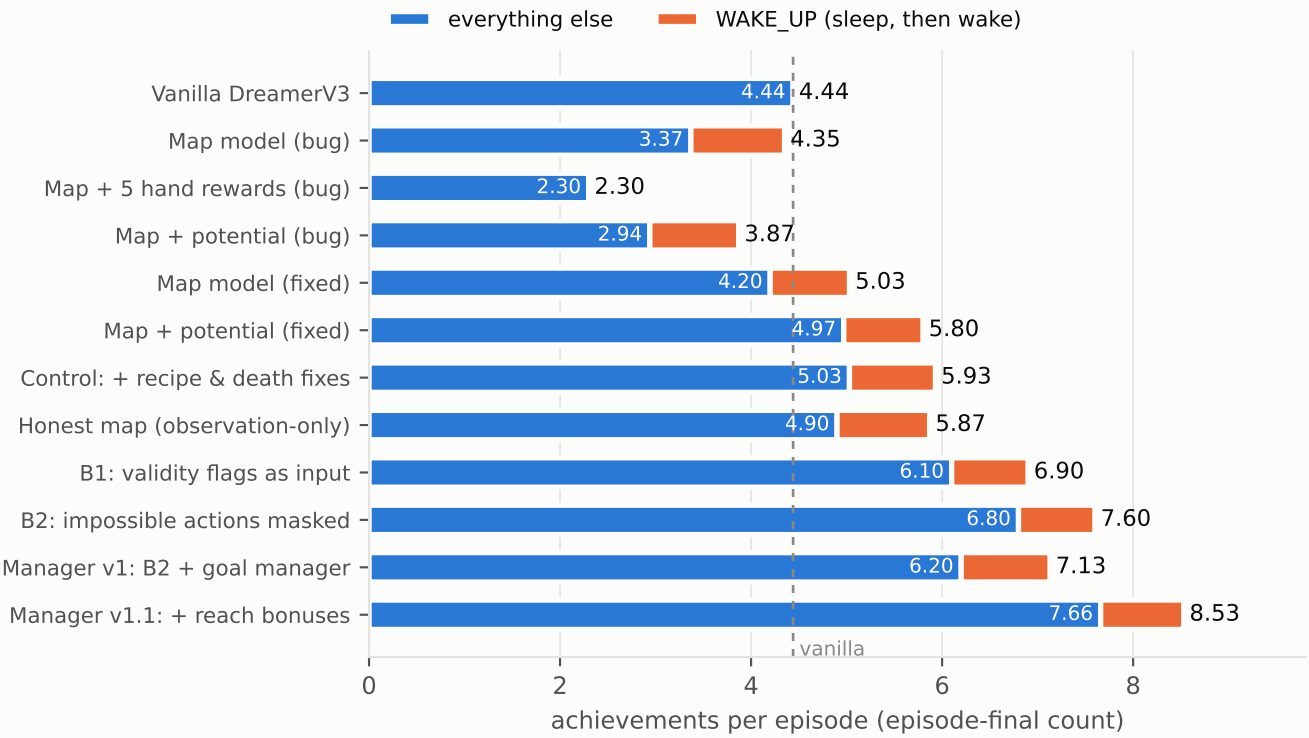
Survival is still unsolved: lifespans 243-311 steps against 261-284 for a random policy, and thirst is the first meter to run out in over half of v1.1's deaths. In normalized return v1.1 reaches 3.4% of the 226-point maximum (B2 3.0%), against 7.09% for the best published 1M-step agent (ITC). The masking and the manager's goals use game knowledge at test time, which the published agents do not.

run	eps	ach.	w/o wake	life	table	pickaxe	sword	stone	drink
Vanilla DreamerV3	25	4.44	4.44	262	64%	24%	0%	12%	80%
Map model (bug)	60	4.35	3.37	294	35%	0%	0%	0%	43%
Map + 5 hand rewards (bug)	50	2.30	2.30	263	0%	0%	0%	0%	44%
Map + potential (bug)	30	3.87	2.94	277	13%	0%	0%	0%	37%
Map model (fixed)	30	5.03	4.20	264	63%	7%	0%	0%	70%
Map + potential (fixed)	30	5.80	4.97	278	90%	30%	10%	20%	60%
Control: + recipe & death fixes	30	5.93	5.03	268	83%	0%	60%	0%	67%
Honest map (observation-only)	30	5.87	4.90	243	87%	20%	20%	7%	83%
B1: validity flags as input	30	6.90	6.10	311	87%	50%	30%	33%	80%
B2: impossible actions masked	30	7.60	6.80	246	87%	53%	20%	43%	70%
Manager v1: B2 + goal manager	30	7.13	6.20	254	73%	57%	23%	47%	63%
Manager v1.1: + reach bonuses	30	8.53	7.66	259	93%	73%	33%	73%	57%

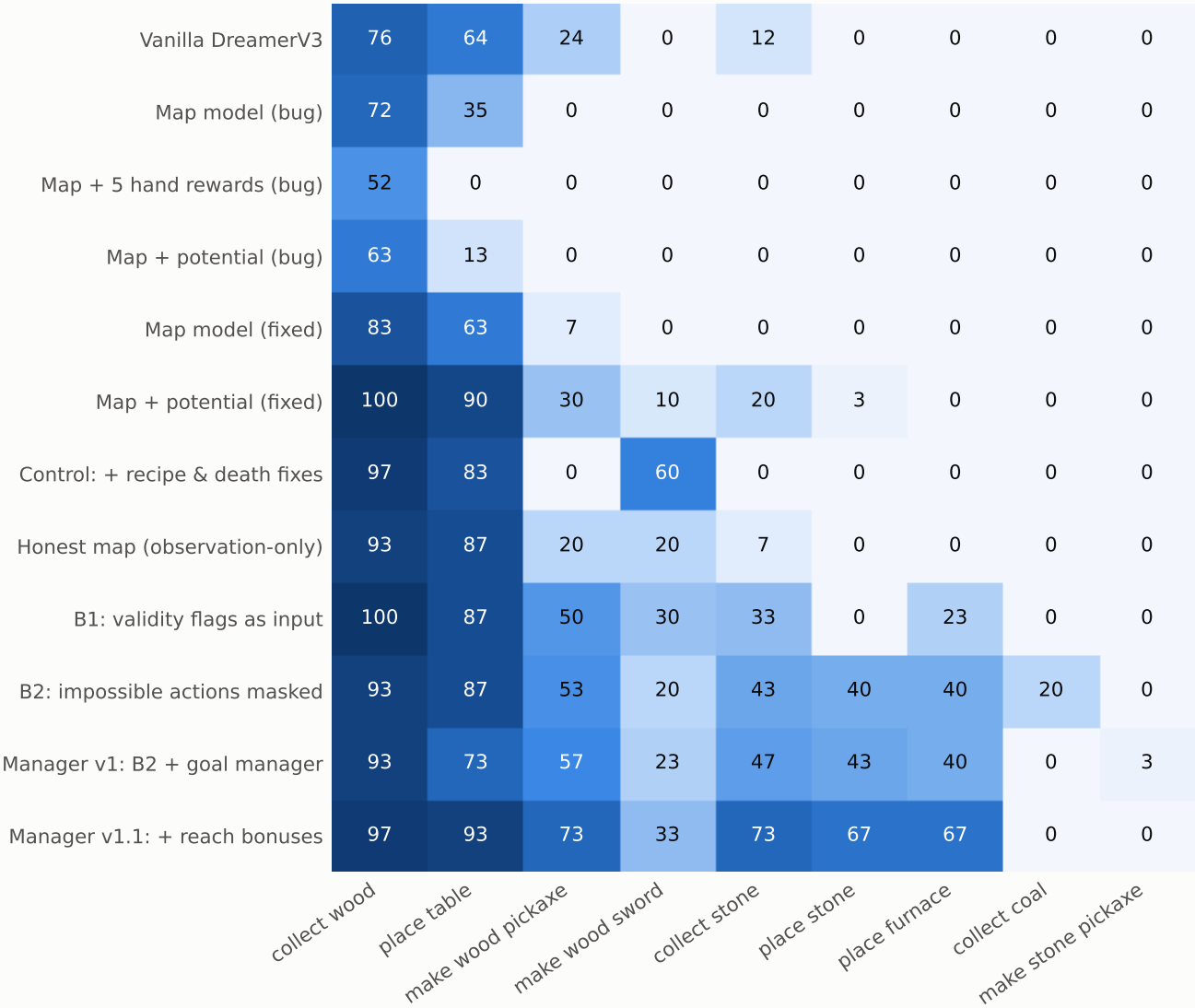
eps = evaluation episodes. ach. = mean episode-final achievements; w/o wake = the same minus WAKE_UP. life = mean lifespan in steps. Rates are % of episodes. 'bug' = imag_shift True, which corrupted the policy gradient (fixed 2026-09-20). map_survive was re-evaluated on 2026-09-28 and reproduced its original 2.30. One training seed per run, so the run-to-run spread is unknown. 'Map + potential (fixed)' is the old 5.80 record; 'Control' adds the recipe and death fixes with the privileged map; 'Honest map' has the same fixes with observation-only targets. 'B1' and 'B2' are experiment B on the honest-map configuration: validity flags as an encoder input, and impossible actions masked in training. 'mgr v1' and 'mgr v1.1' are B2 plus the two-level agent (docs/design-manager.md): v1 pays the manager the game reward only; v1.1 also pays both levels when the goal is reached and masks goals that already hold.

Sleep inflates the map runs; masking and the manager are real p

Mean episode-final achievements per run, split into WAKE_UP and the rest; unlock rate of the early tech-tree rungs



Unlock rate (% of episodes), early tech tree in dependency order



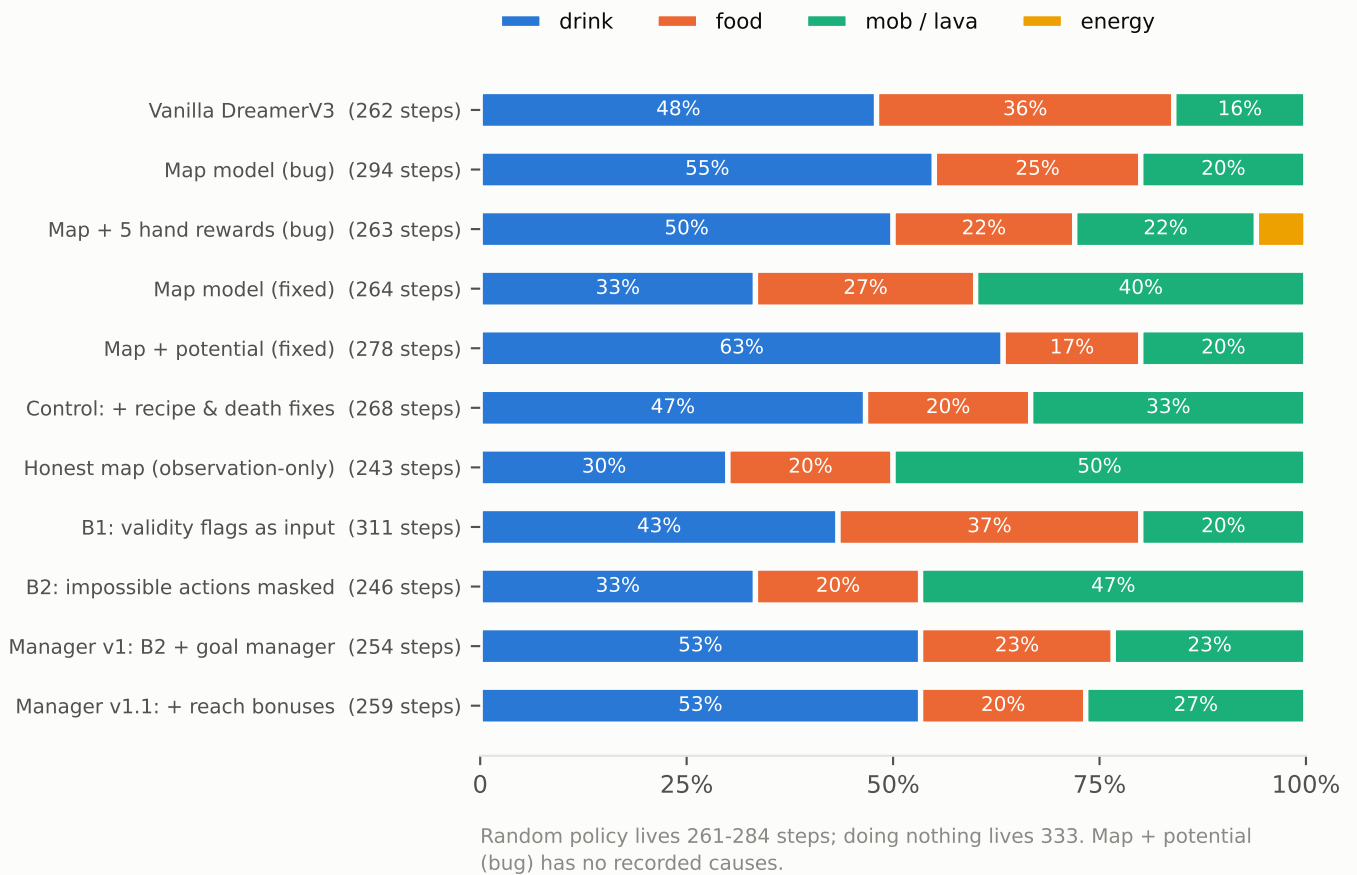
Every achievement any run unlocked

% of evaluation episodes in which each achievement fired. The other 47 of 67 were never unlocked by any run.

[illegible]

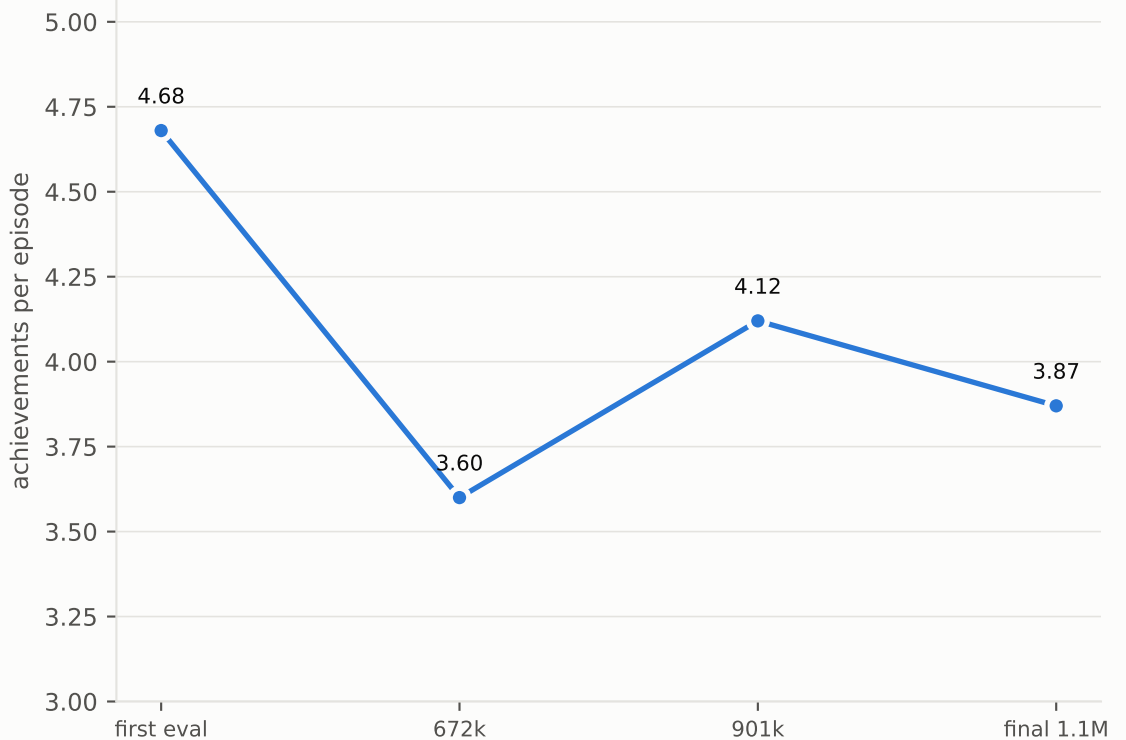
Survival never moved

Cause of death (share of evaluation episodes) and lifespan; and how much one run moves between its own checkpoints



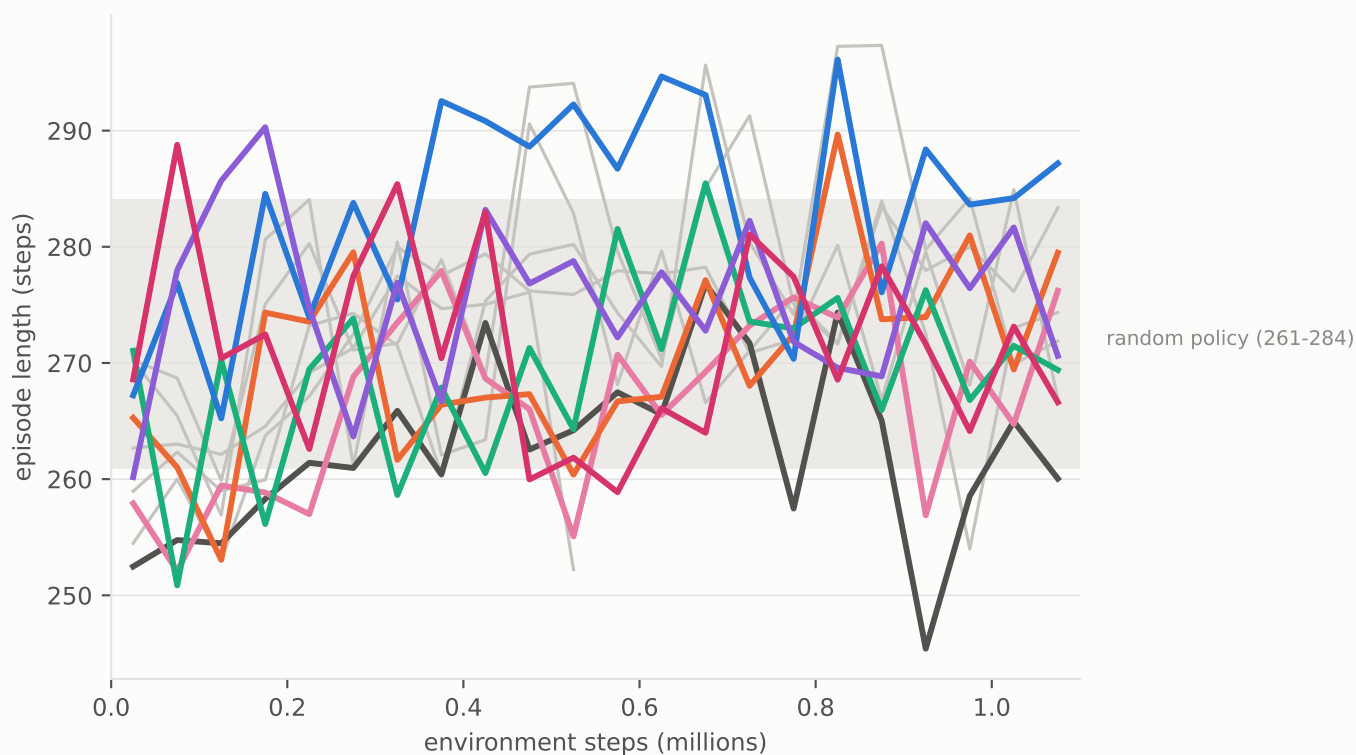
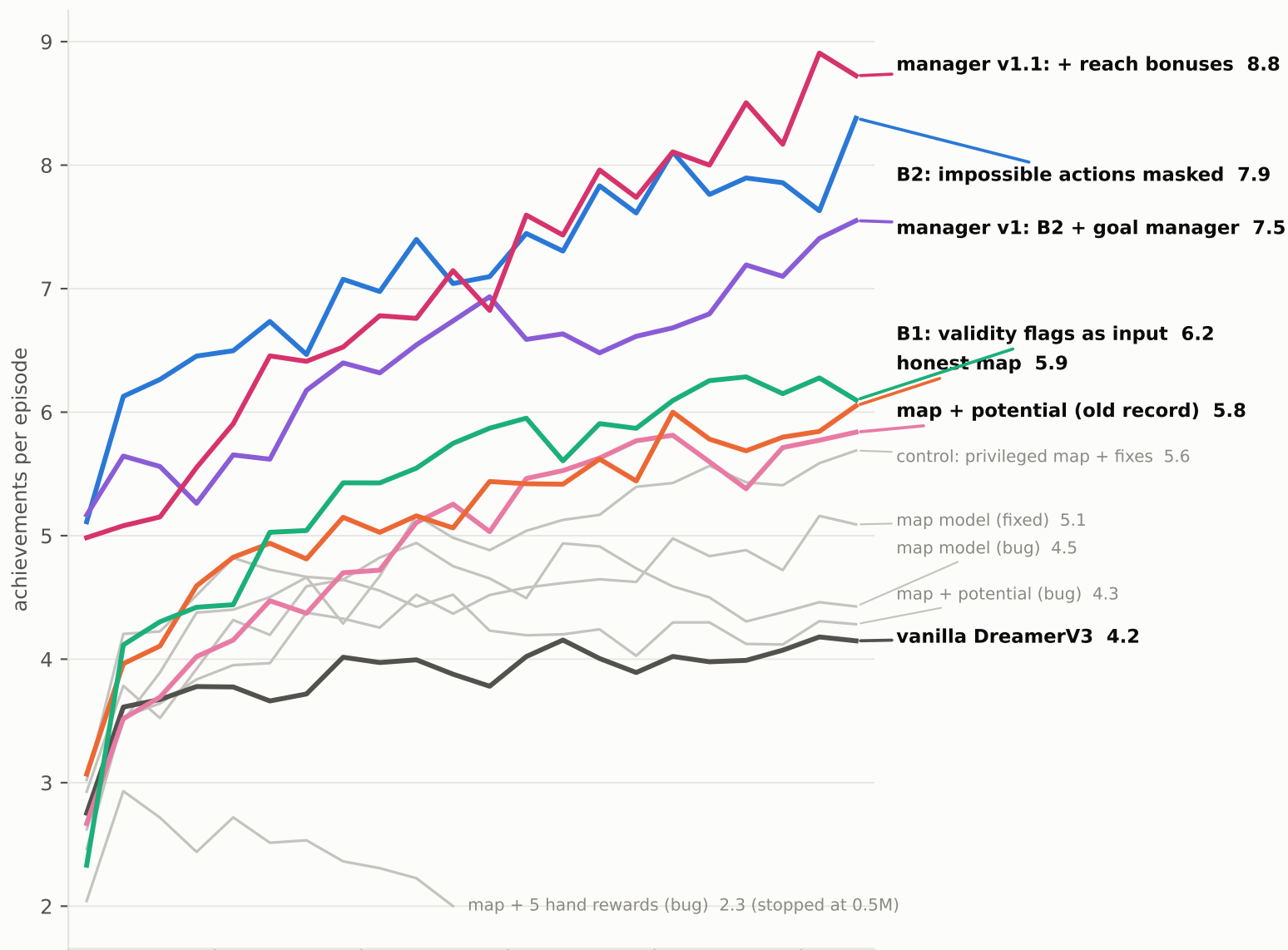
One run, four checkpoints: Map + potential (bug)

The same training run, scored at four points (25-30 episodes each), moves by 1.08 achievements. That is the scale of noise any comparison between single runs has to beat.



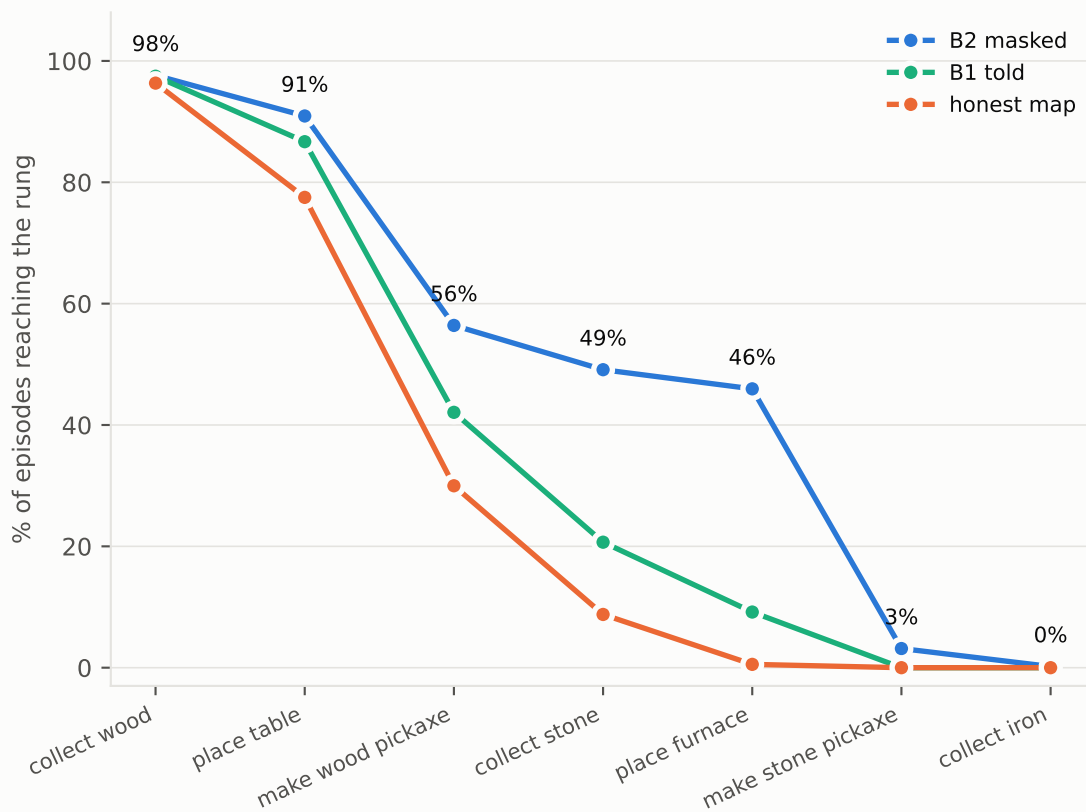
Training curves: every run

Rebuilt from each run's replay; 50k-step bins; value = mean of the last 100k steps; one seed per run.



Where B2 falls short

Last 150k training steps of each run (~500 episodes), decoded from replay by tools/diagnose_run.py



1. The stone pickaxe is the wall. B2 reaches the furnace in 46% of episodes but makes the stone pickaxe in 3%. Of the 249 episodes that collected stone, 73% held wood and stone at the same time, but only 22% ever stood next to a table while holding both, and 6% made the pickaxe. Two gaps: it rarely brings the ingredients back to a table (logistics), and when it does the chance lasts one step -- the single log usually goes into the wood sword or it walks off (measured in docs/b2-casebook.pdf).
2. Table to wood pickaxe leaks. 38% of the episodes that place a table never make the wood pickaxe. At a table holding wood it crafts every time, but every one of these episodes placed the table with its last two logs, and later logs go into another table rather than back to the first (docs/b2-casebook.pdf).
3. Survival. Training-window lifespan 285 (median 275) against 261-284 for a random policy. Thirst is still the first meter to run out in 45% of deaths and hunger in 24%: Crafttax pays for the first drink only, so topping up is never rewarded. A hostile mob is adjacent at 61% of deaths; it dies in lava in 4% (the others: under 1%), the price of exploring further. Deaths in its sleep are rare (5%).
4. Wasted presses. 43% of presses still do nothing (experiment A): DO with nothing in front (26%), NOOP and walking into walls. Masking only removes the special actions; these basic ones are never masked.

What the results say, and what would settle the rest

0. A manager that learns to choose beats one that does not

v1 and v1.1 differ only in how goals pay. In v1 most goals cannot be reached in 8 steps, so the actor ignored them and the manager stayed uniform (entropy $\ln 13$ throughout); it scored below B2 (7.13). v1.1 pays the actor +1 when its goal is reached and the manager +0.5 the first time a goal it set is reached; its manager left uniform within 15k steps, sharpened to entropy 1.65, and the goals it set were reached 0.31 times per imagined rollout by the end. It overtook B2 at about 550k steps and finished 0.9 achievements ahead. One seed per run, so the size of the gap is uncertain; the direction matches the manager metrics.

1. Masking impossible actions is the largest gain so far

B2 7.60 vs honest map 5.87, same model and reward. Experiment A on its final checkpoint shows why: buttons that sat at exactly 0% in the states where they would work now get pressed there (furnace 3.2% over 599 states, placing stone 2.4%, placing a table 75%, the wood pickaxe 100%), and every no-effect press is now a basic action (DO, NOOP, a blocked move). Once a useless press can no longer teach the agent that a button is bad, it presses that button when it works.

2. Telling helps, masking helps more

B1 (flags as input) reaches 6.90: both wood crafts survive in one run for the first time and the furnace comes off zero, but placing stone stays at 0.0% across 452 chances and about 6% of presses still go to impossible buttons. The information helps; removing the suppressing gradient helps more. Masking changes the action space relative to published Craftax baselines, so it is reported as a separate arm.

3. The next wall is sequencing, not suppression

In B2 the stone pickaxe was craftable in only 5 of 5,167 evaluation steps: the agent mines stone but rarely returns to a table holding both wood and stone. That gates iron and every 3-5-point achievement after it. It is a planning problem, so the next experiment should target multi-step sequences (experiment D, return-then-explore, or a sub-goal signal), not the action distribution.

4. Survival is still unsolved

Lifespans stay at or below the random band, except B1 (311, mean pulled up by a few long episodes; median 265). B2 dies earliest (246) and half of its deaths are mobs or lava: the more it does, the more it is exposed.

5. What to run next

Queued: manager v1.2 (one two-headed critic shared by both actors, goals held until reached), v1.3 (v1.2 plus survival goals and a survival potential, aimed at the thirteenth deaths) and B3 (B2 with the action mask learned instead of transcribed from the rules, so no game rules at test time). Then the best manager version with B3's learned mask, and second seeds of the leaders, since every comparison here is single-seed.